SQL Technical Assessment

Topic: Employee Performance and Project Management System

Difficulty: Intermediate
Total Marks: 25
Duration: 60 Minutes

Problem Statement

A software company wants to develop an Employee Performance and Project Management System using MySQL.

The company maintains employee details, department information, salary details, manager information, and project assignments. The management team wants to analyze employee performance, salary distribution, project budgets, and department-wise performance.

You are assigned to develop the database and generate the required reports using SQL.

Your task is to create the required database and tables, insert suitable sample records, and write SQL queries to solve the following business requirements:

1. Generate an employee report displaying employee names, department names, salary details, joining dates, and years of experience.

2. Identify employees based on their salary range and classify them into salary categories such as Low, Medium, and High.

3. Generate a department-wise report showing the total salary, average salary, highest salary, and number of employees. Display only departments whose average salary is greater than ₹40,000.

4. Display employee names along with their respective manager names and identify employees who do not have a manager.

5. Generate a report showing employee and project combinations, including employees who are not assigned to any project and projects that do not have any employees.

6. Identify employees whose salary is greater than the company's average salary and employees who earn more than the average salary of their respective departments.

7. Find employees who work in the same department as a given employee and employees whose salary is greater than the average salary of the IT department.

8. Identify the second-highest-paid employee in the company and the highest-paid employee in each department.

9. Find employees who are assigned to projects with a budget greater than the average project budget.

10. Identify employees who have worked on projects with more hours than the average project working hours.

11. Generate a report showing each employee's total number of assigned projects and identify employees who are assigned to more projects than the average number of projects per employee.

12. Display all possible employee and project combinations for project planning and resource allocation.

Student Instructions: Create suitable tables with primary key and foreign key constraints, insert at least 5 records into each table, and write the required SQL queries to generate the reports.